

Date: June 1, 2026
Location: Bengaluru, India

To:
The Editorial Board and Editor-in-Chief
IEEE Transactions on Services Computing

Subject: Submission of Research Manuscript for Peer Review

Dear Editor-in-Chief and Members of the Editorial Board,

We submit our research manuscript titled “**Real-Time Shill-Bidding Detection via Online Bid-Graph Analytics in Live Auction Platforms**” for consideration for publication in your journal.

This work addresses shill bidding and collusion in live auctions. While prior literature focuses on offline, post-hoc batch analytics operating after an auction closes, this paper introduces an in-process sliding-window directed outbid graph permitting sub-millisecond streaming feature extraction and logistic regression classification during active bidding. The platform extracts five streaming metrics – including bid frequency, response latency, price increment ratio, seller co-occurrence, and outbid reciprocity – to feed a lightweight machine learning engine. Additionally, the system incorporates cryptographic sealed-bid commitments via a SHA-256 commit-reveal protocol and resolves critical database lock contention. By integrating an asynchronous Redis Stream sequencer, transactional throughput scales substantially, preserving server performance during high-concurrency event loops.

In connection with this submission, we declare the following:

1. **Originality:** This manuscript represents original research. It has not been published previously and is not under consideration for publication elsewhere.
2. **AI Disclosure:** The use of AI assistance in refining code and drafting sections is disclosed in the Acknowledgements section of the manuscript.
3. **Author Approval:** All authors (Asha Joseph, Pratham Reet, Nitin A. Rane, Reeshav Sinha, and Om Ji Rao) have read and approved the manuscript for submission.

Thank you for your time and consideration of our manuscript.

Sincerely,

Pratham Reet
Corresponding Author
Department of Computer Science and Engineering
New Horizon College of Engineering
Bengaluru, India
Email: prathamreet@gmail.com